Drake vs Kendrick Rhyme Scheme Analysis pt. 2

How does lyrical complexity in hip hop relate to fan sentiment and engagement in online communities, specifically on Reddit and YouTube? On top of this, can I apply more metrics to draw deeper conclusions relating to the artist's songs?

I will look at Drake and Kendrick Lamar, reusing rhyme metrics from Project 1 and adding a few simple new lyrical complexity metrics (syllables per line, stress patterns).

In [ ]:
```
!pip install google-api-python-client
!pip install emoji
```

In [1]:
```
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sb
import nltk
from nltk.corpus import cmudict
nltk.download("cmudict") # dictionary for rhyme scheme analysis
from dotenv import load_dotenv
import requests
import re
import sqlite3
from googleapiclient.discovery import build
import urllib.parse
from googleapiclient.errors import HttpError # debug yt error
from nltk.sentiment.vader import SentimentIntensityAnalyzer
nltk.download("vader_lexicon")
```

```
[nltk_data] Downloading package cmudict to
[nltk_data]     /Users/mattabatangle/nltk_data...
[nltk_data]   Package cmudict is already up-to-date!
[nltk_data] Downloading package vader_lexicon to
[nltk_data]     /Users/mattabatangle/nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!
```

Out[1]:
```
True
```

In [2]:
```
load_dotenv("3510.env")
```

Out[2]:
```
True
```

I am going to get started by creating my sqlite DB.

In [3]:
```
dbConnection = sqlite3.connect("Project2.db")
curse = dbConnection.cursor()
```

In [4]:
```
# table for lyric metrics per song
curse.execute("""
    CREATE TABLE IF NOT EXISTS 'songs' (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        song_title TEXT NOT NULL,
        artist TEXT NOT NULL,
        album TEXT,
        rhyme_density REAL,
        rhyme_scheme_variety REAL,
        internal_rhyme_rate REAL,
        avg_syllables_per_line REAL,
        syllable_variance REAL,
        stress_variability REAL,
        word_count INTEGER,
        complexity REAL
    );""")
```

Out[4]:
```
<sqlite3.Cursor at 0x120a07b40>
```

In [5]:
```
# table for reddit posts and comments
curse.execute("""
    CREATE TABLE IF NOT EXISTS reddit_posts (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        reddit_id TEXT UNIQUE,
        author TEXT,
        subreddit TEXT,
        published_at TEXT,
        score INTEGER,
        text TEXT,
        artist_tag TEXT,
        sentiment REAL
    );""")
```

Out[5]:
```
<sqlite3.Cursor at 0x120a07b40>
```

In [6]:
```
# table for YouTube comments
curse.execute("""
    CREATE TABLE IF NOT EXISTS youtube_comments (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        comment_id TEXT UNIQUE,
        author TEXT,
        video_id TEXT,
        published_at TEXT,
        like_count INTEGER,
        text TEXT,
        artist_tag TEXT,
        sentiment REAL
    );""")
```

Out[6]:
```
<sqlite3.Cursor at 0x120a07b40>
```

In [7]:
```
dbConnection.commit()
```

In [8]:
```
songDF = pd.read_csv("Project2Data.csv") # load in data from project 1
# (I just added one line at the bottom of project 1 saving these specific co
songDF.head()
```

Out[8]:

	track_name	artist	album_name	word_count	rhyme_density	complexity
0	tv off (feat. lefty gunplay)	Kendrick Lamar	GNX	770	0.929504	715.718016
1	HUMBLE.	Kendrick Lamar	DAMN.	598	0.892562	533.752066
2	Swimming Pools (Drank)	Kendrick Lamar	Swimming Pools (Drank)	859	0.928654	797.713464
3	LOVE. FEAT. ZACARI.	Kendrick Lamar	DAMN.	488	0.947242	462.254197
4	PRIDE.	Kendrick Lamar	DAMN.	504	0.880626	443.835616

In [9]:
```
# add song data from projec 1 to sqlite db
for i, row in songDF.iterrows():
        curse.execute("""
        INSERT INTO songs (
            song_title,
            artist,
            album,
            avg_syllables_per_line,
            syllable_variance,
            stress_variability,
            word_count,
            rhyme_density,
            complexity
        )
        VALUES (?, ?, ?, NULL, NULL, NULL, ?, ?, ?);
    """, (
        row["track_name"],
        row["artist"],
        row["album_name"],
        row["word_count"],
        row["rhyme_density"],
        row["complexity"]
    )) # right now I am using NULL as a placeholder for all the values I wil
```

In [10]:
```
dbConnection.commit()
```

Next, I want to add a few more stats and go a little deeper with the lyrics (hinted at above)

In [11]:
```
lyricsDF = pd.read_csv("project2Lyrics.csv") # load in lyrics from project 1
```

In [12]:
```
lyricsDF.head()
```

Out[12]:

	track_name	artist	lyrics
0	tv off (feat. lefty gunplay)	Kendrick Lamar	All I ever wanted was a black Grand National\n...
1	HUMBLE.	Kendrick Lamar	Nobody pray for me\nIt been that day for me\nW...
2	Swimming Pools (Drank)	Kendrick Lamar	Pour up (Drank), head shot (Drank)\nSit down (...
3	LOVE. FEAT. ZACARI.	Kendrick Lamar	Maldición, amor o lujuria\nMaldición, todos no...
4	PRIDE.	Kendrick Lamar	Love's gonna get you killed\nBut pride's gonna...

First, I want to set up CMU dict again and set up syllable counting with a fallback in case the word is not present in the CMU Dict

In [13]:
```
cmu = cmudict.dict()
```

In [14]:
```
def get_cmu_pronunciations(word): # return list of pronunciations from CMU
    word = word.lower()
    return cmu.get(word, []) # or empty list if missing
```

In [15]:
```
def count_syllables_from_cmu(pron): # count syllables in one CMU pronunciati
    count = 0
    for phone in pron: # loop over phonemes in pronunciation
        if any(char.isdigit() for char in phone): # check if any character i
            count += 1 # count it
    return count
```

In [16]:
```
def estimate_syllables_fallback(word): # if word not in dict (highly unlikel
    word = word.lower() # lower
    word = re.sub(r"[^a-z]", "", word) # remove punctuation
    if not word:
        return 0
    groups = re.findall(r"[aeiouy]+", word) # counting vowel groups
    return max(1, len(groups)) # 1 in case word has no vowels
```

In [17]:
```
def count_syllables_in_word(word):
    prons = get_cmu_pronunciations(word) # earlier function
    if prons:
        # use first pronunciation
        return count_syllables_from_cmu(prons[0])
    else:
        return estimate_syllables_fallback(word)
```

Next, I want to look at stress and variability in wording. This is the closest I can get to rhythmic density with just lyrics.

In [18]:
```
def get_stress_sequence_for_line(line): # return a list of stress digits (0,
    stresses = [] # list
```

```
    words = re.findall(r"[A-Za-z']+", line) # extract words
    for w in words:
        prons = get_cmu_pronunciations(w) # call earlier function
        if not prons:
            continue # if CMU has no entry, skip the word entirely
        pron = prons[0]
        for phone in pron:
            # phones with digits are vowels with stress markers (used last v
            digits = [char for char in phone if char.isdigit()] # extract nu
            if digits: # check phone has digit
                stresses.append(int(digits[0]))
    return stresses
```

In [19]:
```
def compute_stress_variability(stress_seq): # Compute fraction of syllable p
    if len(stress_seq) < 2:
        return 0.0 # cannot measure variation with fewer than two syllables
    changes = 0 # counter for stress changes
    for i in range(1, len(stress_seq)): # check if the current stress differ
        if stress_seq[i] != stress_seq[i - 1]:
            changes += 1
    # The denominator is the number of syllable pairs (no matter if it switc
    return changes / (len(stress_seq) - 1)
```

Ok, now I am going to combine the previous too sections of functions into one giant function to clean things up.

In [20]:
```
def compute_line_metrics(lyrics_text): # compute above metrics for input lyr
    lines = [ln.strip() for ln in lyrics_text.split("\n")] # split on new li
    # keep only non empty lines with at least one letter (real lines)
    lines = [ln for ln in lines if re.search(r"[A-Za-z]", ln)]

    if not lines:
        return 0.0, 0.0, 0.0

    syllable_counts = []
    all_stresses = []

    for line in lines: # line by line
        words = re.findall(r"[A-Za-z']+", line) # extract words
        if not words:
            continue # just skip

        # syllables
        syllables_in_line = sum(count_syllables_in_word(w) for w in words) #
        syllable_counts.append(syllables_in_line) # add to total count

        # stress pattern for this line
        stresses = get_stress_sequence_for_line(line) # call prev function
        all_stresses.extend(stresses) # add list to list

    if not syllable_counts: # if somehow it never got any syllable counts, r
        avg_syll = 0.0
        var_syll = 0.0
    else:
        # compute the average number of syllables per line
```

```
        avg_syll = float(np.mean(syllable_counts))
        # compute the variance in syllables per line
        var_syll = float(np.var(syllable_counts)) # higher variance means sc

    stress_var = compute_stress_variability(all_stresses) # over whole song

    return avg_syll, var_syll, stress_var
```

Time to perform and add these calculations to the DB

In [21]:
```
for i, row in lyricsDF.iterrows(): # iterat through and assign new vals
    title = row["track_name"]
    artist = row["artist"]
    lyrics_text = row["lyrics"]

    avg_syll, var_syll, stress_var = compute_line_metrics(lyrics_text)

    curse.execute("""
        UPDATE songs
        SET avg_syllables_per_line = ?,
            syllable_variance = ?,
            stress_variability = ?
        WHERE song_title = ? AND artist = ?;
    """, (avg_syll, var_syll, stress_var, title, artist))
```

In [22]:
```
dbConnection.commit()
```

Due to Reddit API restrictions I focused the community analysis on YouTube.

In [23]:
```
# I am going to remove the reddit table here instead of deleting the initial
curse.execute("DROP TABLE IF EXISTS reddit_posts;")
dbConnection.commit()
```

In [24]:
```
ytKey = os.getenv("YOUTUBE_KEY")
```

In [25]:
```
# API setup
youtube = build("youtube", "v3", developerKey=ytKey)
```

Start by searching for the best video for a song

In [26]:
```
def search_video_for_song(artist, track_name, max_results=1): # simple call
    query = f"{artist} {track_name} official audio"

    request = youtube.search().list(part="snippet",q=query, type="video", ma
    response = request.execute()

    items = response.get("items", [])
    if not items:
        return None

    # take first result
    video_id = items[0]["id"]["videoId"]
    return video_id
```

In [27]:
```
curse.execute("SELECT song_title, artist FROM songs LIMIT 1;")
song_title, artist = curse.fetchone()
```

In [28]:
```
print("Testing:", artist, "-", song_title)
print("Video ID:", search_video_for_song(artist, song_title))
```

```
Testing: Kendrick Lamar - tv off (feat. lefty gunplay)
Video ID: U8F5G5wR1mk
```

If you search the ID on youtube the first result SHOULD be the song (was for me).

In [29]:
```
def insert_youtube_comment(comment_id, author, video_id, published_at, like_
    curse.execute("""
        INSERT OR IGNORE INTO youtube_comments (
            comment_id,
            author,
            video_id,
            published_at,
            like_count,
            text,
            artist_tag,
            sentiment
        ) VALUES (?, ?, ?, ?, ?, ?, ?, NULL);""", # sentiment will be filled
                (comment_id, author, video_id, published_at, like_count, tex
```

Next I want to actually scrape the comment data.

In [30]:
```
def fetch_comments_for_video(video_id, artist_tag, max_comments=200):
    fetched = 0 # counting how many saved comments
    next_page_token = None # used for iteration through yt page results (par

    while fetched < max_comments: # not maxed out
        remaining = max_comments - fetched
        page_size = min(100, remaining) # whichever lower
        # req taking in comment snippet
        request = youtube.commentThreads().list(part="snippet", videoId=vide

        try: # added in later debug
            response = request.execute()

        except HttpError as e:
            # if the video has comments disabled (added this later due to co
            if e.resp.status == 403 and "commentsDisabled" in str(e):
                print(f"Comments disabled for video {video_id}. Skipping.")
                return 0 # nothing fetched
            # otherwise rethrow
            raise e

        for item in response['items']:
            snippet = item['snippet']['topLevelComment']['snippet'] # take c

            comment_id = item['id']
            author = snippet.get('authorDisplayName')
            text = snippet.get('textOriginal', '') # default blank
            like_count = snippet.get('likeCount', 0) # default 0
```

```
            published_at = snippet.get('publishedAt')
            # call prev function to insert into the DB
            insert_youtube_comment(comment_id=comment_id, author=author, vic

            fetched += 1 # count comment saved
            if fetched >= max_comments: # if max: quit
                break

        next_page_token = response.get("nextPageToken") # next page (similar
        if not next_page_token: # if no more pages (no more comments) break
            break

    dbConnection.commit()
    print(f"Fetched {fetched} comments for video {video_id} ({artist_tag})")
```

Since this is going to be a ton of data I want to test it with two songs first

In [31]:
```
curse.execute("""
    SELECT song_title, artist
    FROM songs
    WHERE artist = 'Drake'
    LIMIT 1;""")
drake_songs = curse.fetchall()
curse.execute("""
    SELECT song_title, artist
    FROM songs
    WHERE artist = 'Kendrick Lamar'
    LIMIT 1;""")
kendrick_songs = curse.fetchall()
print(drake_songs, kendrick_songs)
```

```
[('One Dance', 'Drake')] [('tv off (feat. lefty gunplay)', 'Kendrick Lama
r')]
```

In [32]:
```
for title, artist in drake_songs + kendrick_songs: # only two for test
    print(f"Processing: {artist} - {title}")
    video_id = search_video_for_song(artist, title)
    if not video_id:
        print("No video found, skipping.")
        continue # same as how I did it with Spotify API -- skip if no resul

    print("Found video:", video_id)
    fetch_comments_for_video(video_id=video_id, artist_tag=artist,max_commer
```

```
Processing: Drake - One Dance
Found video: FOqKN-ouAUE
Fetched 100 comments for video FOqKN-ouAUE (Drake)
Processing: Kendrick Lamar - tv off (feat. lefty gunplay)
Found video: U8F5G5wR1mk
Fetched 100 comments for video U8F5G5wR1mk (Kendrick Lamar)
```

In [33]:
```
curse.execute("""
    SELECT artist_tag, COUNT(*)
    FROM youtube_comments
    GROUP BY artist_tag;""")
print(curse.fetchall())
```

```
[('Drake', 100), ('Kendrick Lamar', 100)]
```

In [34]:
```
len(songDF[songDF["artist"] == "Drake"])
```

Out[34]: 34

In [35]:
```
len(songDF[songDF["artist"] == "Kendrick Lamar"])
```

Out[35]: 23

Need the above numbers to know limit in SQL query.

In [36]:
```
curse.execute("""
    SELECT song_title, artist
    FROM songs
    WHERE artist = 'Drake'
    LIMIT 34;""")
drake_songs = curse.fetchall()
curse.execute("""
    SELECT song_title, artist
    FROM songs
    WHERE artist = 'Kendrick Lamar'
    LIMIT 23;""")
kendrick_songs = curse.fetchall()
```

In [37]:
```
for title, artist in drake_songs + kendrick_songs:
    print(f"Processing: {artist} - {title}")
    video_id = search_video_for_song(artist, title)
    if not video_id:
        print("No video found, skipping.")
        continue # same as how I did it with Spotify API -- skip if no resul

    print("Found video:", video_id)
    fetch_comments_for_video(video_id, artist_tag=artist, max_comments=200)
```

```
Processing: Drake - One Dance
Found video: ki0Ocze98U8
Fetched 200 comments for video ki0Ocze98U8 (Drake)
Processing: Drake - Passionfruit
Found video: EgfsXTOn_pI
Comments disabled for video EgfsXTOn_pI. Skipping.
Processing: Drake - Headlines
Found video: Sn3SUnL44w4
Comments disabled for video Sn3SUnL44w4. Skipping.
Processing: Drake - Not You Too (feat. Chris Brown)
Found video: ZX_mvoY_Hg0
Fetched 200 comments for video ZX_mvoY_Hg0 (Drake)
Processing: Drake - NOKIA
Found video: RDH71p3LgWM
Comments disabled for video RDH71p3LgWM. Skipping.
Processing: Drake - God's Plan
Found video: m1a_GqJf02M
Comments disabled for video m1a_GqJf02M. Skipping.
Processing: Drake - Which One (feat. Central Cee)
Found video: 9-dEHfSCZUQ
Comments disabled for video 9-dEHfSCZUQ. Skipping.
Processing: Drake - Teenage Fever
Found video: e8HtwsnuTIw
Comments disabled for video e8HtwsnuTIw. Skipping.
Processing: Drake - DIE TRYING
Found video: CiV9BHZCaKQ
Fetched 200 comments for video CiV9BHZCaKQ (Drake)
Processing: Drake - 9
Found video: q50SwIodCwg
Comments disabled for video q50SwIodCwg. Skipping.
Processing: Drake - Jimmy Cooks (feat. 21 Savage)
Found video: V7UgPHjN9qE
Fetched 200 comments for video V7UgPHjN9qE (Drake)
Processing: Drake - Hotline Bling
Found video: zt6aRKpf9T4
Fetched 200 comments for video zt6aRKpf9T4 (Drake)
Processing: Drake - What Did I Miss?
Found video: 85stUoBMbcY
Comments disabled for video 85stUoBMbcY. Skipping.
Processing: Drake - Yebba’s Heartbreak
Found video: 9rlW2rUzyn0
Fetched 200 comments for video 9rlW2rUzyn0 (Drake)
Processing: Drake - SOMEBODY LOVES ME
Found video: htToA3wcFl4
Fetched 200 comments for video htToA3wcFl4 (Drake)
Processing: Drake - Trust Issues
Found video: 87plLWRWNZA
Comments disabled for video 87plLWRWNZA. Skipping.
Processing: Drake - Best I Ever Had
Found video: Zfp3KfYH0xA
Comments disabled for video Zfp3KfYH0xA. Skipping.
Processing: Drake - Chicago Freestyle (feat. Giveon)
Found video: p9pf5EyOgcs
Fetched 200 comments for video p9pf5EyOgcs (Drake)
Processing: Drake - Practice
Found video: JUrDOWj9RUw
```

```
Comments disabled for video JUrDOWj9RUw. Skipping.
Processing: Drake - Rich Baby Daddy (feat. Sexyy Red & SZA)
Found video: F7o0upORtCw
Fetched 200 comments for video F7o0upORtCw (Drake)
Processing: Drake - Marvins Room
Found video: JDb3ZZD4bA0
Comments disabled for video JDb3ZZD4bA0. Skipping.
Processing: Drake - Do Not Disturb
Found video: vVd4T5NxLgI
Comments disabled for video vVd4T5NxLgI. Skipping.
Processing: Drake - Over My Dead Body
Found video: PMk8L9FNqnY
Comments disabled for video PMk8L9FNqnY. Skipping.
Processing: Drake - Nonstop
Found video: QVqS3tB8OtE
Comments disabled for video QVqS3tB8OtE. Skipping.
Processing: Drake - Take Care
Found video: 5bBcMt4mS2o
Comments disabled for video 5bBcMt4mS2o. Skipping.
Processing: Drake - Nice For What
Found video: 1Jx4Dv269uE
Comments disabled for video 1Jx4Dv269uE. Skipping.
Processing: Drake - Rich Flex
Found video: I4DjHHVHWAE
Fetched 200 comments for video I4DjHHVHWAE (Drake)
Processing: Drake - Hold On, We're Going Home
Found video: GxgqpCdOKak
Fetched 200 comments for video GxgqpCdOKak (Drake)
Processing: Drake - Live For
Found video: GLr0usOC2_k
Fetched 200 comments for video GLr0usOC2_k (Drake)
Processing: Drake - Miss Me
Found video: 18DjQ1knr6I
Comments disabled for video 18DjQ1knr6I. Skipping.
Processing: Drake - LOYAL (feat. Drake)
Found video: FL7icWyial0
Fetched 200 comments for video FL7icWyial0 (Drake)
Processing: Drake - TWIST & TURN (feat. Drake & PARTYNEXTDOOR)
Found video: 5N8YRren5Z4
Fetched 200 comments for video 5N8YRren5Z4 (Drake)
Processing: Drake - She Will
Found video: 502ECjkTedc
Comments disabled for video 502ECjkTedc. Skipping.
Processing: Drake - My Love (feat. Drake) - Remix
Found video: VI3LrB0-21I
Comments disabled for video VI3LrB0-21I. Skipping.
Processing: Kendrick Lamar - tv off (feat. lefty gunplay)
Found video: U8F5G5wR1mk
Fetched 200 comments for video U8F5G5wR1mk (Kendrick Lamar)
Processing: Kendrick Lamar - HUMBLE.
Found video: ov4WobPqoSA
Comments disabled for video ov4WobPqoSA. Skipping.
Processing: Kendrick Lamar - Swimming Pools (Drank)
Found video: X0sVhnP15z8
Comments disabled for video X0sVhnP15z8. Skipping.
Processing: Kendrick Lamar - LOVE. FEAT. ZACARI.
```

```
Found video: ox7RsX1Ee34
Fetched 200 comments for video ox7RsX1Ee34 (Kendrick Lamar)
Processing: Kendrick Lamar - PRIDE.
Found video: pRGmFiEyr0A
Comments disabled for video pRGmFiEyr0A. Skipping.
Processing: Kendrick Lamar - squabble up
Found video: 7QyDL3zQ-2I
Comments disabled for video 7QyDL3zQ-2I. Skipping.
Processing: Kendrick Lamar - Alright
Found video: JocAXINz-YE
Comments disabled for video JocAXINz-YE. Skipping.
Processing: Kendrick Lamar - Money Trees
Found video: bBNpSXAYteM
Comments disabled for video bBNpSXAYteM. Skipping.
Processing: Kendrick Lamar - DNA.
Found video: ue4xoNqdc2I
Comments disabled for video ue4xoNqdc2I. Skipping.
Processing: Kendrick Lamar - peekaboo (feat. azchike)
Found video: cbHkzwa0QmM
Fetched 200 comments for video cbHkzwa0QmM (Kendrick Lamar)
Processing: Kendrick Lamar - Count Me Out
Found video: 6nTcdw7bVdc
Fetched 200 comments for video 6nTcdw7bVdc (Kendrick Lamar)
Processing: Kendrick Lamar - Die Hard
Found video: Lx3MGrafykU
Fetched 200 comments for video Lx3MGrafykU (Kendrick Lamar)
Processing: Kendrick Lamar - euphoria
Found video: NPqDIwWMtxg
Fetched 200 comments for video NPqDIwWMtxg (Kendrick Lamar)
Processing: Kendrick Lamar - LOYALTY. FEAT. RIHANNA.
Found video: sN8H2ypmzlA
Comments disabled for video sN8H2ypmzlA. Skipping.
Processing: Kendrick Lamar - United In Grief
Found video: tvNSXS4x9nc
Fetched 200 comments for video tvNSXS4x9nc (Kendrick Lamar)
Processing: Kendrick Lamar - King Kunta
Found video: AC4bb9Q9-04
Comments disabled for video AC4bb9Q9-04. Skipping.
Processing: Kendrick Lamar - N95
Found video: XEV4x7xEprw
Fetched 200 comments for video XEV4x7xEprw (Kendrick Lamar)
Processing: Kendrick Lamar - Push Thru
Found video: gVMfnvBAsSc
Comments disabled for video gVMfnvBAsSc. Skipping.
Processing: Kendrick Lamar - Malcolm X
Found video: U8F5G5wR1mk
Fetched 200 comments for video U8F5G5wR1mk (Kendrick Lamar)
Processing: Kendrick Lamar - Bitch, Don’t Kill My Vibe - Remix
Found video: JZ5-Soe3NJ0
Fetched 200 comments for video JZ5-Soe3NJ0 (Kendrick Lamar)
Processing: Kendrick Lamar - Two Presidents
Found video: vyOMKFHT6eE
Comments disabled for video vyOMKFHT6eE. Skipping.
Processing: Kendrick Lamar - BLOOD.
Found video: hcnNvy_svTE
Comments disabled for video hcnNvy_svTE. Skipping.
```

```
Processing: Kendrick Lamar - YAH.
Found video: kL9QqLugfLU
Comments disabled for video kL9QqLugfLU. Skipping.
```

Looks like there are a ton of different reasons Youtube API blocks comments (record labels, stricter guidlines with API, etc.)

On the bright side I think this is enough data to continue on

In [38]:
```
# bring in comments to pandas df
commentsDF = pd.read_sql_query("SELECT * FROM youtube_comments", dbConnectic
commentsDF.head()
```

Out[38]:

	id	comment_id	author	video_id	pul
0	1	UgxC9OkGPPrsmUZslKB4AaABAg	@jilliannotice8852	FOqKN-ouAUE	10
1	2	UgxjWZNe1TI43Y1h-7t4AaABAg	@9020966883	FOqKN-ouAUE	10
2	3	UgwNeYsUTWE9nab-Wst4AaABAg	@notzy0	FOqKN-ouAUE	09
3	4	Ugwtp60ApYzhdlEN6ah4AaABAg	@AlexaVargasHolayomellamoAlexaG	FOqKN-ouAUE	05
4	5	UgxH7eK0OxynQYVwjBI4AaABAg	@fatimaammarjaffery1774	FOqKN-ouAUE	04

In [39]:
```
vaderSIA = SentimentIntensityAnalyzer()
```

In [40]:
```
def compute_sentiment(text): # quick function (same use as class)
    return vaderSIA.polarity_scores(text)['compound']
```

In [41]:
```
commentsDF['sentiment'] = commentsDF['text'].apply(compute_sentiment) # add
```

Now I am going to save these back to the DB

In [42]:
```
for i, row in commentsDF.iterrows():
    curse.execute("""
        UPDATE youtube_comments
        SET sentiment = ?
        WHERE id = ?""",
                  (row['sentiment'], row['id'])) # use key
```

In [43]:
```
dbConnection.commit()
```

In [44]:
```
# writing over both as this will be the data I work with from here
songDF = pd.read_sql_query("SELECT * FROM songs", dbConnection)
commentsDF = pd.read_sql_query("SELECT * FROM youtube_comments", dbConnectic
```

In [45]:
```
songDF.head()
```

Out[45]:

	id	song_title	artist	album	rhyme_density	rhyme_scheme_variety	internal
0	1	tv off (feat. lefty gunplay)	Kendrick Lamar	GNX	0.929504	None	
1	2	HUMBLE.	Kendrick Lamar	DAMN.	0.892562	None	
2	3	Swimming Pools (Drank)	Kendrick Lamar	Swimming Pools (Drank)	0.928654	None	
3	4	LOVE. FEAT. ZACARI.	Kendrick Lamar	DAMN.	0.947242	None	
4	5	PRIDE.	Kendrick Lamar	DAMN.	0.880626	None	

In [46]:
```
commentsDF.head()
```

Out[46]:

	id	comment_id	author	video_id	pul
0	1	UgxC9OkGPPrsmUZslKB4AaABAg	@jilliannotice8852	FOqKN-ouAUE	10
1	2	UgxjWZNe1TI43Y1h-7t4AaABAg	@9020966883	FOqKN-ouAUE	10
2	3	UgwNeYsUTWE9nab-Wst4AaABAg	@notzy0	FOqKN-ouAUE	09
3	4	Ugwtp60ApYzhdlEN6ah4AaABAg	@AlexaVargasHolayomellamoAlexaG	FOqKN-ouAUE	05T
4	5	UgxH7eK0OxynQYVwjBl4AaABAg	@fatimaammarjaffery1774	FOqKN-ouAUE	04

I want to add in some more stats for comments with the data alrady gathered

In [47]:
```
def word_count(text):
    return len(text.split())
```

In [48]:
```
import emoji
def emoji_count(text):
    return sum(1 for ch in text if ch in emoji.EMOJI_DATA) # add 1 everytime
```

In [49]:
```
commentsDF['word_count'] = commentsDF['text'].apply(word_count)
commentsDF['emoji_count'] = commentsDF['text'].apply(emoji_count)
```

In [50]:
```
comments_stats = commentsDF.groupby('artist_tag').agg({ # group by artist ar
    'sentiment': 'mean',
    'word_count': 'mean',
    'emoji_count': 'mean',
    'like_count': 'mean',
    'comment_id': 'count'}).rename(columns={'comment_id': 'num_comments'})
# take count of unique comment id (i think youtube gives a unique id even if
```

In [51]:
```
comments_stats # above comment was right
```

Out[51]:

artist_tag	sentiment	word_count	emoji_count	like_count	num_comments
Drake	0.093567	12.453793	1.241379	4.295517	2900
Kendrick Lamar	0.077690	17.725000	1.610556	2.368333	1800

There are already some clear differences between the two artists' comment sections. Kendrick Lamar's videos show a substantially higher average word count, which likely reflects the more layered themes and narratives in his music. Listeners appear to engage with his videos in a more analytical way, producing longer and more detailed comments. Drake's comments seem to be shorter but slightly more positive on average. His videos also show a much higher average like count, which may reflect the broader mainstream reach of his music and the fact that his music often invites lighter, more universally agreeable reactions rather than deep thematic discussion. The emoji count also follows this pattern, signalling stronger opinions related to Kendrick's content.

Next I want to look at song level stats

In [52]:

```
song_stats = songDF.groupby('artist').agg({ # group by artist and aggregate
    'rhyme_density': 'mean',
    'word_count': 'mean',
    'complexity': 'mean',
    'avg_syllables_per_line': 'mean',
    'syllable_variance': 'mean',
    'stress_variability': 'mean'})
```

In [53]:

```
song_stats
```

Out[53]:

artist	rhyme_density	word_count	complexity	avg_syllables_per_line	syllable_va
Drake	0.917521	544.147059	501.950562	10.081031	15.4
Kendrick Lamar	0.899463	675.130435	614.407159	10.938610	18.7

In my last project I already looked at rhyme density, word count, and complexity. Now, I have introduced three more metrics to look into. These measures reveal that Kendrick Lamar consistently uses slightly more syllables per line, which suggests denser or faster flows. His syllable variance is also noticeably higher, which indicates more irregular line structures and greater variation in pacing across verses. Rhythm really cannot be directly extracted from text alone, but these indicators together can point to more dynamic and complex rhythmic patterning in Kendrick's writing. His higher stress variability further supports this by showing greater fluctuation in where emphasis falls within lines. It is a bit of a leap, but these structural complexities may help explain why Kendrick's comment sections contain longer and more detailed responses. Listeners

might engage more deeply with lyrics that present richer rhythmic and linguistic complexity.

In [54]:
```
combined_stats = song_stats.join( comments_stats, how='inner', lsuffix='_sor
# since there is two "word_count" columns, the suffix will specify which is
# just like SQL Inner Join
combined_stats
```

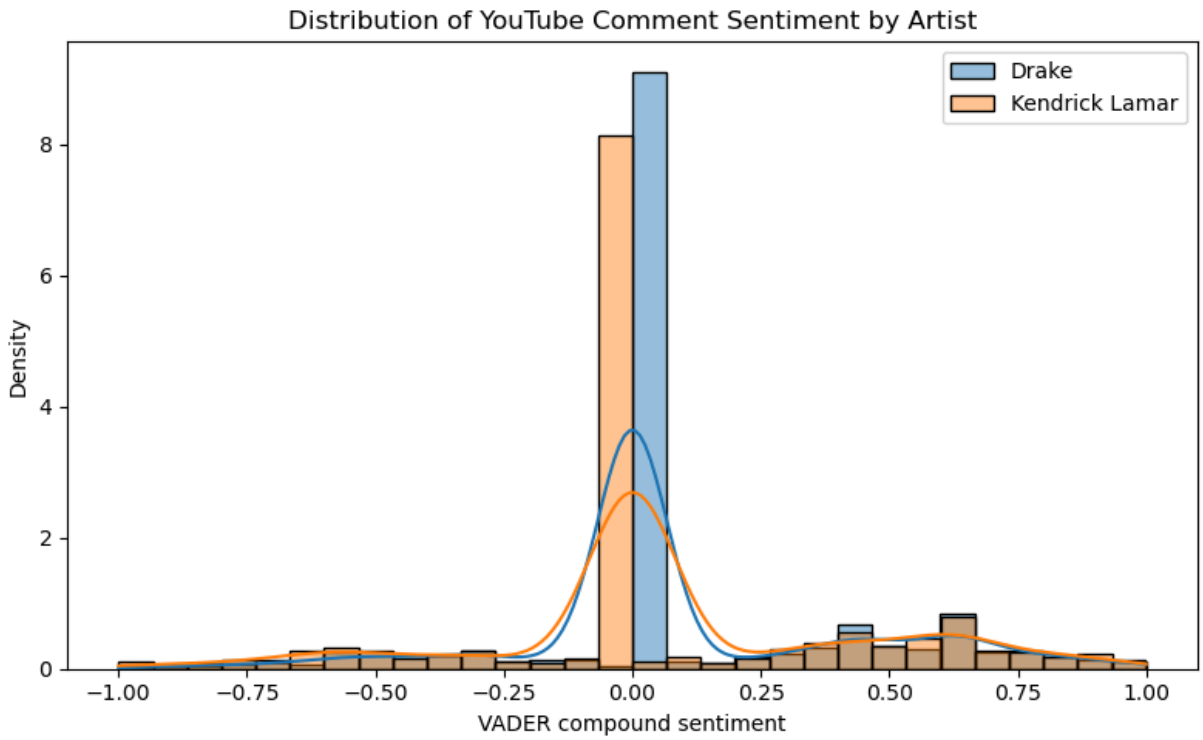
Out[54]:

artist	rhyme_density	word_count_song	complexity	avg_syllables_per_line	syllab
Drake	0.917521	544.147059	501.950562	10.081031	
Kendrick Lamar	0.899463	675.130435	614.407159	10.938610	

Nice and clean combined statistic display

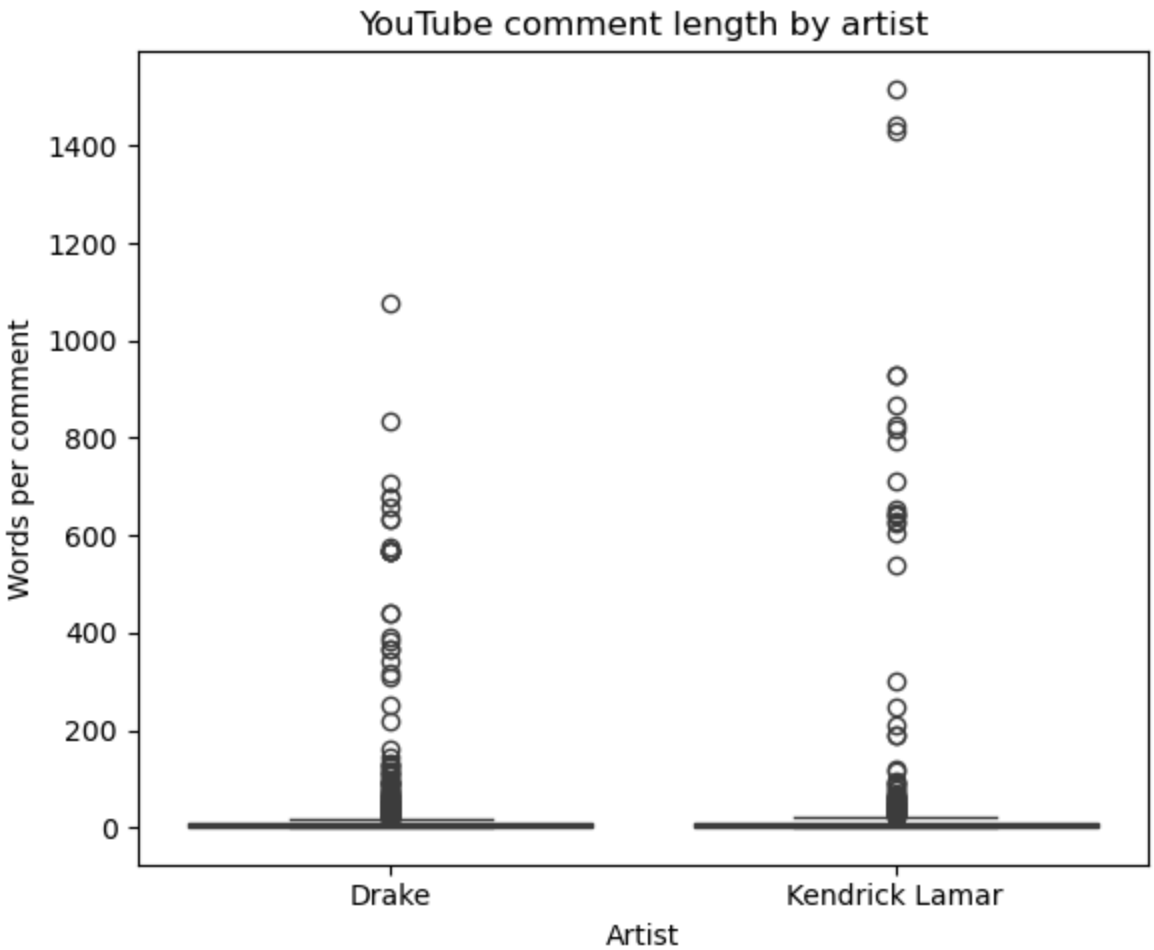
For my first visualization, I want to display the key metrics by artist (Drake vs Kendrick)

In [55]:
```
plt.figure(figsize=(8, 5))
# Drake
sb.histplot(data=commentsDF[commentsDF["artist_tag"] == "Drake"], x="sentime

# Kendrick
sb.histplot(data=commentsDF[commentsDF["artist_tag"] == "Kendrick Lamar"], x

plt.xlabel("VADER compound sentiment")
plt.ylabel("Density")
plt.title("Distribution of YouTube Comment Sentiment by Artist")
plt.legend()
plt.tight_layout()
plt.show()
```



The sentiment distributions for Drake and Kendrick Lamar are both heavily centered near zero. This indicates that most comments are neutral or only mildly positive. As shown earlier in the aggregate statistics, the mean sentiment for both artists sits just above zero. One interesting detail is the noticeable secondary bump around the 0.6 range. This suggests that a sizeable subset of comments are distinctly positive rather than just neutral, which may reflect moments where fans are reacting to a specific song, lyric, or cultural moment with stronger enthusiasm.

In [56]:
```
plt.figure(figsize=(6, 5))
sb.boxplot(data=commentsDF, x="artist_tag", y="word_count")
plt.xlabel("Artist")
plt.ylabel("Words per comment")
plt.title("YouTube comment length by artist")
plt.tight_layout()
plt.show()
```



Comments on Kendrick Lamar videos tend to be longer on average and show a wider spread in length. This supports the earlier summary table where his mean comment word count was higher. Fans appear to write more detailed or reflective responses to his songs, while Drake's comment sections are shorter and more compact overall.

For my last visualization I was planning on connecting the two forms of analysis (lyrical + fan sentiment) but between API rate limits and a good amount of missing comment sections it feels more structured to keep them seperate.

Instead, I want to visualize the new lyrical components I calculated earlier throughout both artists careers. This will be interesting to compared with my temporal visualzation in project 1, looking at just rhyme density over time.

To avoid issues chaning the starting CSV, I will just import a new version from project 1 with just release year, artist, and song title.

In [57]:
```
yearDF = pd.read_csv("project2ReleaseYears.csv")
yearDF
```

Out[57]:

	track_name	artist	release_year
0	tv off (feat. lefty gunplay)	Kendrick Lamar	2024
1	HUMBLE.	Kendrick Lamar	2017
2	Swimming Pools (Drank)	Kendrick Lamar	2012
3	LOVE. FEAT. ZACARI.	Kendrick Lamar	2017
4	PRIDE.	Kendrick Lamar	2017
5	squabble up	Kendrick Lamar	2024
6	Alright	Kendrick Lamar	2015
7	Money Trees	Kendrick Lamar	2012
8	DNA.	Kendrick Lamar	2017
9	peekaboo (feat. azchike)	Kendrick Lamar	2024
10	Count Me Out	Kendrick Lamar	2022
11	Die Hard	Kendrick Lamar	2022
12	euphoria	Kendrick Lamar	2024
13	LOYALTY. FEAT. RIHANNA.	Kendrick Lamar	2017
14	United In Grief	Kendrick Lamar	2022
15	King Kunta	Kendrick Lamar	2015
16	N95	Kendrick Lamar	2022
17	Push Thru	Kendrick Lamar	2012
18	Malcolm X	Kendrick Lamar	2021
19	Bitch, Don’t Kill My Vibe - Remix	Kendrick Lamar	2013
20	Two Presidents	Kendrick Lamar	2013
21	BLOOD.	Kendrick Lamar	2017
22	YAH.	Kendrick Lamar	2017
23	One Dance	Drake	2016
24	Passionfruit	Drake	2017
25	Headlines	Drake	2011
26	Not You Too (feat. Chris Brown)	Drake	2020
27	NOKIA	Drake	2025
28	God's Plan	Drake	2018
29	Which One (feat. Central Cee)	Drake	2025
30	Teenage Fever	Drake	2017
31	DIE TRYING	Drake	2025

	track_name	artist	release_year
32	9	Drake	2016
33	Jimmy Cooks (feat. 21 Savage)	Drake	2022
34	Hotline Bling	Drake	2016
35	What Did I Miss?	Drake	2025
36	Yebba's Heartbreak	Drake	2021
37	SOMEBODY LOVES ME	Drake	2025
38	Trust Issues	Drake	2019
39	Best I Ever Had	Drake	2009
40	Chicago Freestyle (feat. Giveon)	Drake	2020
41	Practice	Drake	2011
42	Rich Baby Daddy (feat. Sexyy Red & SZA)	Drake	2023
43	Marvins Room	Drake	2011
44	Do Not Disturb	Drake	2017
45	Over My Dead Body	Drake	2011
46	Nonstop	Drake	2018
47	Take Care	Drake	2011
48	Nice For What	Drake	2018
49	Rich Flex	Drake	2022
50	Hold On, We're Going Home	Drake	2013
51	Live For	Drake	2013
52	Miss Me	Drake	2010
53	LOYAL (feat. Drake)	Drake	2019
54	TWIST & TURN (feat. Drake & PARTYNEXTDOOR)	Drake	2020
55	She Will	Drake	2011
56	My Love (feat. Drake) - Remix	Drake	2016

In [58]:
```
songDF = songDF.rename(columns={"song_title": "track_name"})
```

In [59]:
```
mergedDF = songDF.merge(yearDF, on=["track_name", "artist"], how="inner") #
```

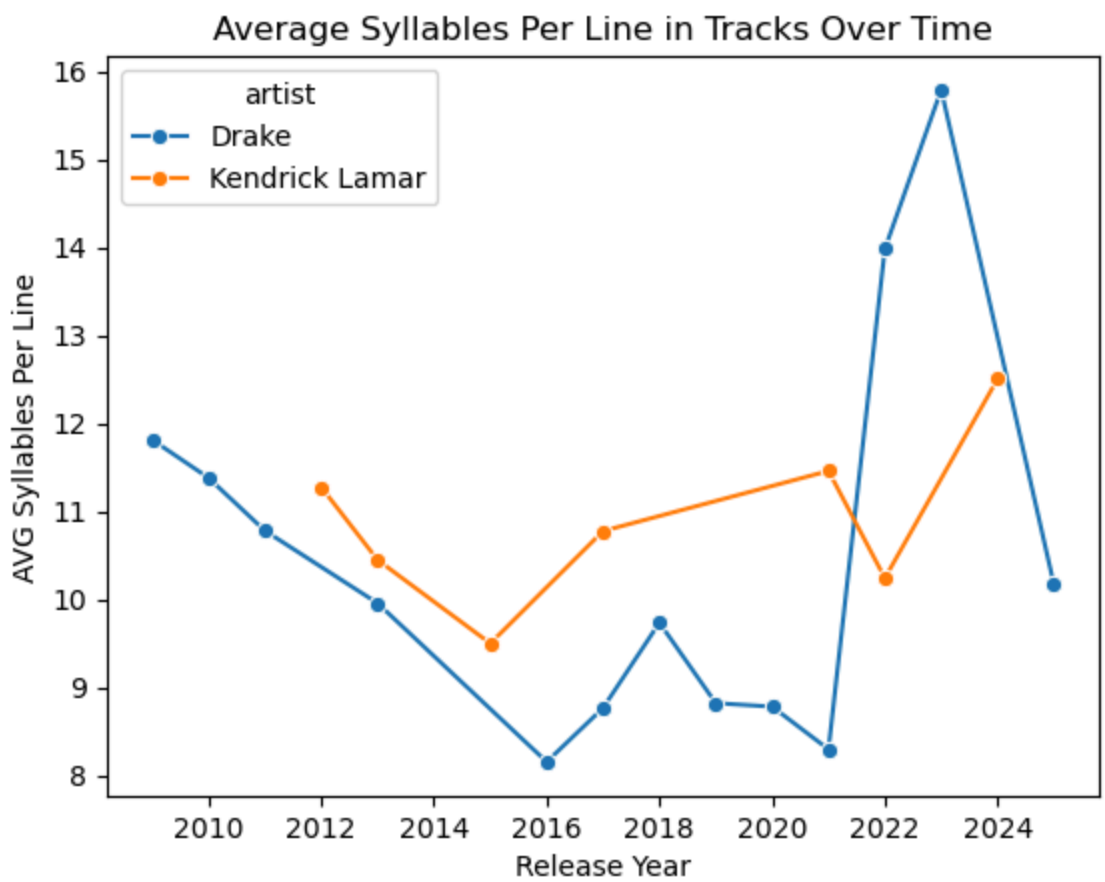
In [60]:
```
mergedDF.head()
```

Out[60]:

	id	track_name	artist	album	rhyme_density	rhyme_scheme_variety	intern
0	1	tv off (feat. lefty gunplay)	Kendrick Lamar	GNX	0.929504	None	
1	2	HUMBLE.	Kendrick Lamar	DAMN.	0.892562	None	
2	3	Swimming Pools (Drank)	Kendrick Lamar	Swimming Pools (Drank)	0.928654	None	
3	4	LOVE. FEAT. ZACARI.	Kendrick Lamar	DAMN.	0.947242	None	
4	5	PRIDE.	Kendrick Lamar	DAMN.	0.880626	None	

In [61]:

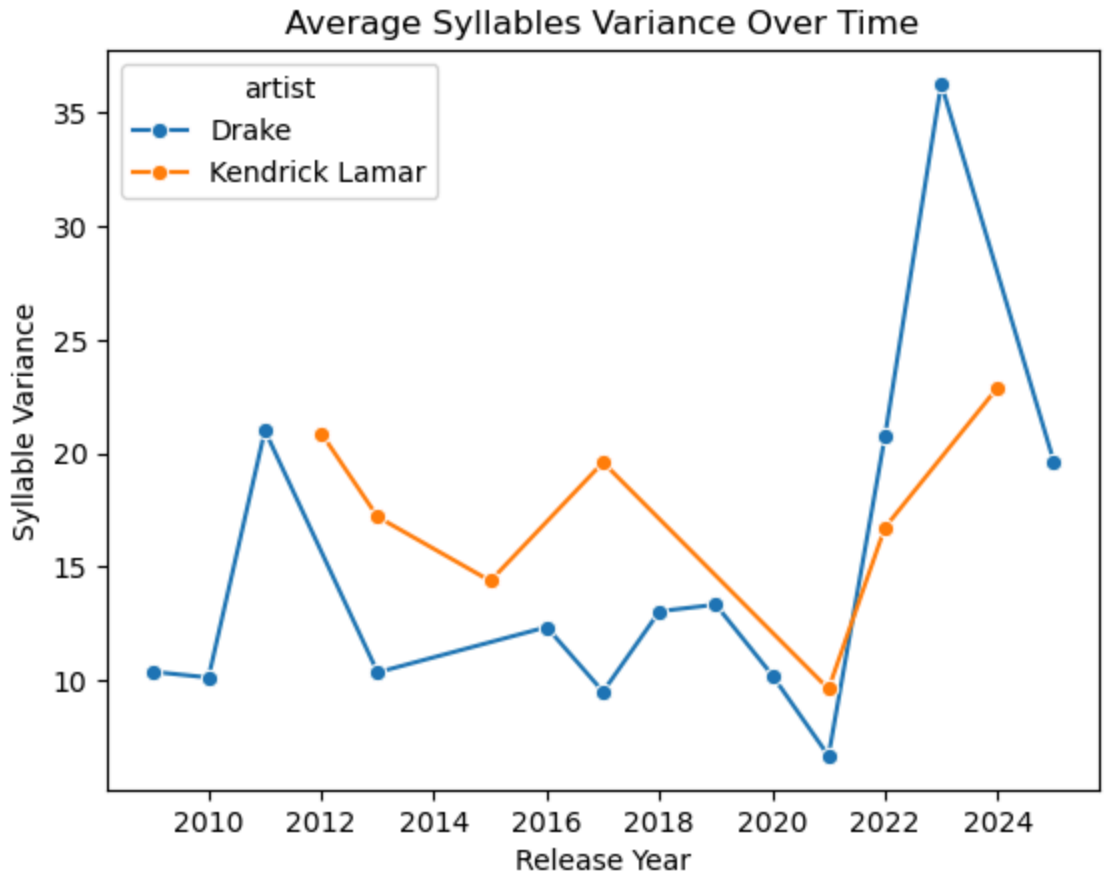
```
yearly = (mergedDF.groupby(["artist", "release_year"])["avg_syllables_per_li
sb.lineplot(data=yearly, x="release_year", y="avg_syllables_per_line", hue="
plt.title("Average Syllables Per Line in Tracks Over Time")
plt.xlabel("Release Year")
plt.ylabel("AVG Syllables Per Line")
plt.show()
```



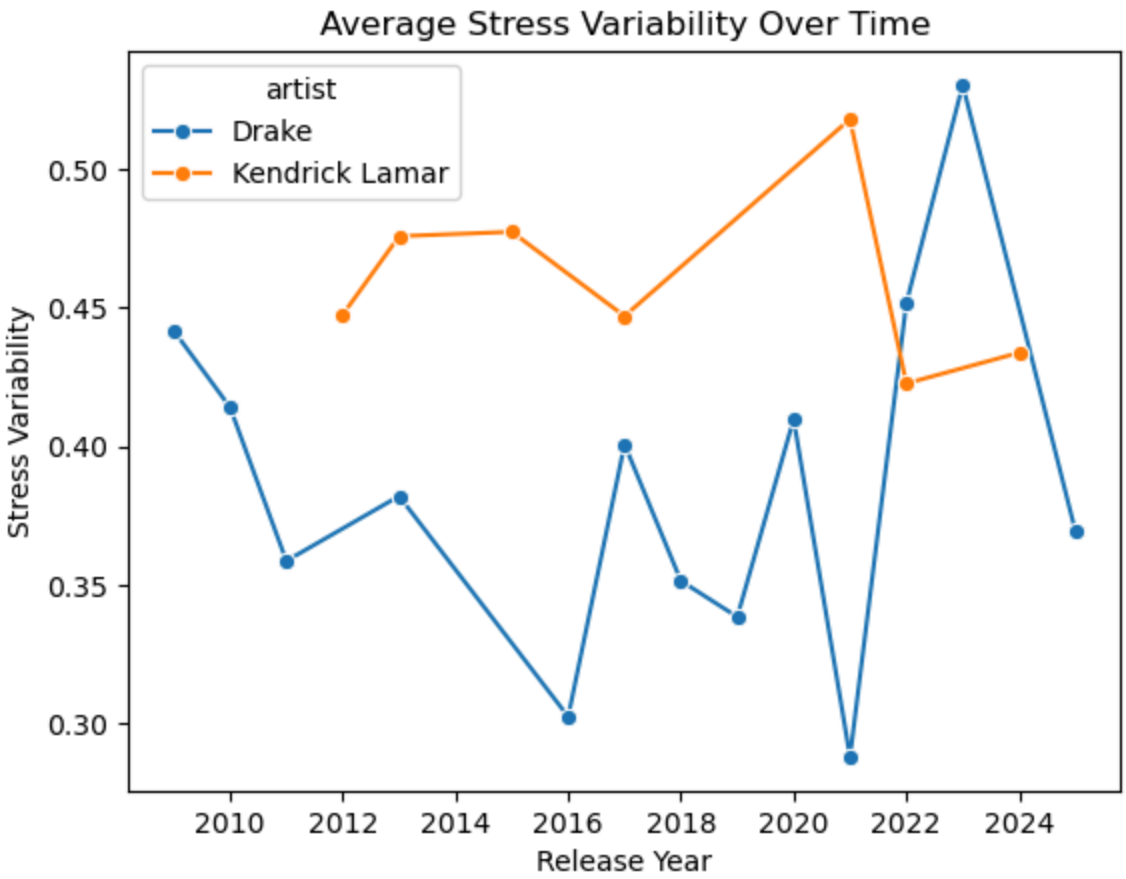
In [62]:

```
yearly = (mergedDF.groupby(["artist", "release_year"])["syllable_variance"].
sb.lineplot(data=yearly, x="release_year", y="syllable_variance", hue="artis
plt.title("Average Syllables Variance Over Time")
```

```
plt.xlabel("Release Year")
plt.ylabel("Syllable Variance")
plt.show()
```



In [63]:
```
yearly = (mergedDF.groupby(["artist", "release_year"])["stress_variability"]
sb.lineplot(data=yearly, x="release_year", y="stress_variability", hue="arti
plt.title("Average Stress Variability Over Time")
plt.xlabel("Release Year")
plt.ylabel("Stress Variability")
plt.show()
```



Across all three metrics, Kendrick Lamar shows a consistently higher, (and more variable), linguistic complexity compared to Drake. Kendrick's fluctuations across album cycles suggest experimentation with pacing, rhythm, and structural density. Drake's trends are smoother and generally lower, which aligns with his emphasis on melody, delivery, and conversational language. The year-to-year variation also shows that neither artist is static. Kendrick's spikes correspond to stylistic pivots between major releases, while Drake's changes show shifts between pop-leaning and rap-leaning projects. These patterns help contextualize my earlier findings about comment behavior and complexity discussions in fan communities.